SAP Fiori — Procure-to-Pay (P2P) Interactive Prototype

Name: Rajarshi Mukherjee | Roll Number: 23052091 | Batch/Program: SAP FIORI (IE)

GitHub Link: https://github.com/rajarshi-29/FioriNex

Live Demo Link: https://rajarshi-29.github.io/FioriNex/

1. Title

SAP Fiori — Procure-to-Pay (P2P) Interactive Prototype
Company: FioriNex Pvt Ltd | Subject: SAP Fiori | Academic Year: 2025–26

2. Problem Statement

FioriNex Pvt Ltd is a mid-size manufacturing and procurement company operating plants across Mumbai and Pune on SAP S/4HANA. The company's procurement team currently relies on traditional SAP GUI transaction codes such as ME51N, ME21N, MIGO, and MIRO to execute the full Procure-to-Pay cycle. These interfaces are complex, non-intuitive, and inaccessible on mobile and tablet devices, leading to frequent errors in purchase requisitions, missed goods receipt postings, and delayed supplier invoice processing.

The organisation requires a modern, role-based SAP Fiori application that simplifies the complete P2P cycle — from raising a Purchase Requisition to releasing supplier payment — through an intuitive tile-based Fiori Launchpad accessible on both desktop and tablet. The solution must follow SAP Fiori design principles using OpenUI5, simulate OData service responses for all five P2P steps, and be deployable without requiring a live SAP backend system.

3. Solution & Features

The solution is an interactive SAP Fiori prototype built using OpenUI5 (the open-source SAPUI5 framework) following the standard MVC architecture. XML

views define the UI, JavaScript controllers handle business logic, and a JSON model simulates SAP OData service responses. A central Fiori Launchpad provides six application tiles covering all five stages of the P2P cycle.

P2P Process Flow covered:

Purchase Requisition (ME51N) → PR Approval (ME28) → Purchase Order (ME21N) → Goods Receipt (MIGO – Movement Type 101) → Invoice Verification / 3-Way Match (MIRO)

Features by screen:

Fiori Launchpad: Tile-based home screen replicating a real SAP Fiori Launchpad. Six GenericTile components display live counters (e.g., "4 Requisitions Open", "2 Pending Approval"). A collapsible P2P process flow banner at the top guides users through all five steps sequentially.

My Purchase Requisitions: Responsive table view of all PRs with live search by item or PR ID, status filter dropdown (Pending / Approved / Rejected), and colour-coded ObjectStatus badges. Tapping any row displays PR details via a toast notification.

Create Purchase Requisition: A structured SimpleForm with required field validation covering item description, quantity (StepInput), unit of measure, plant, vendor, estimated amount, department, required date, and remarks. On submission, the new PR is added dynamically to the JSON model and immediately reflects in the PR list and launchpad counter.

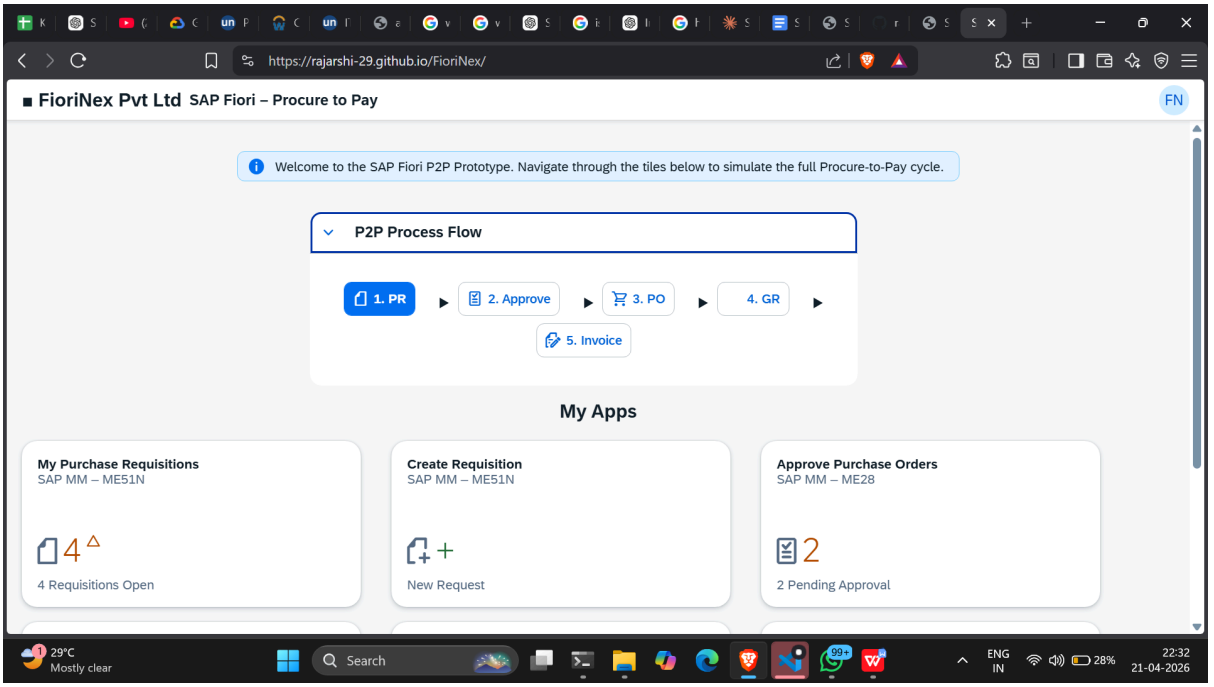
Approve Purchase Orders: ObjectListItem list showing full PR details per item. Each item opens an ActionSheet with individual Approve or Reject options. A toolbar-level "Approve All" button allows bulk approval of all pending PRs with a confirmation dialog.

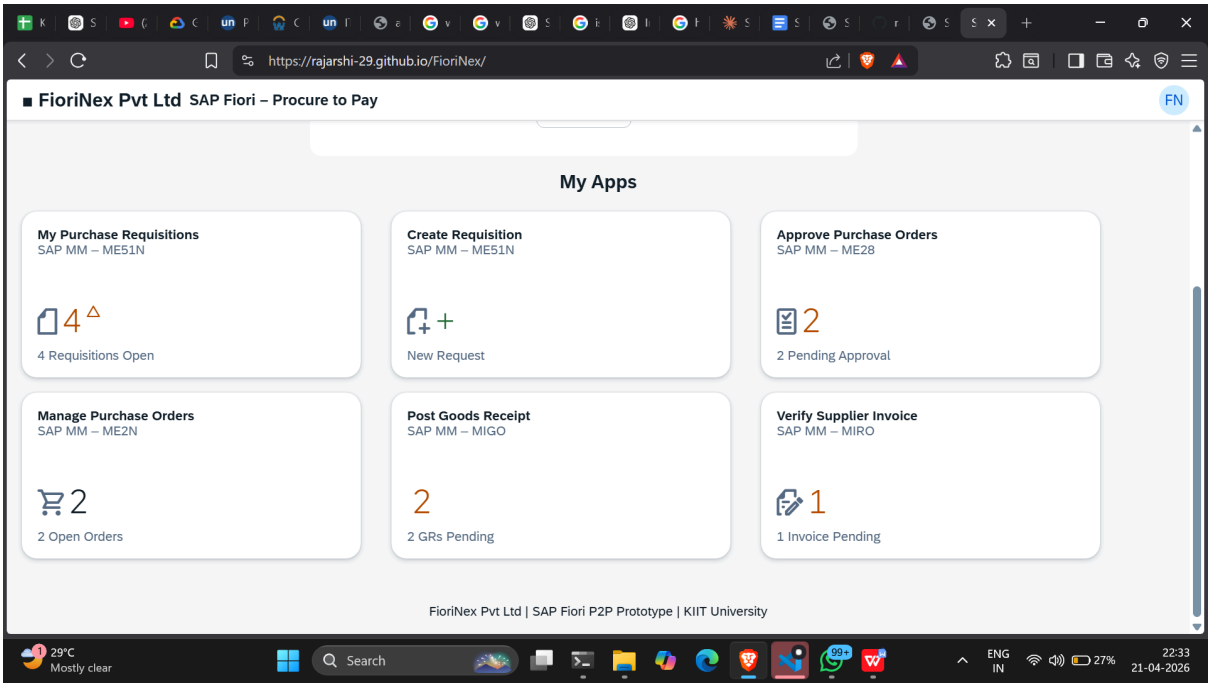
Manage Purchase Orders: Tabular view of all Purchase Orders generated from approved PRs, showing vendor, item, quantity, amount, delivery date, payment terms, and open/closed status.

Post Goods Receipt: Table of pending GRs with StepInput controls for entering received quantity against each PO. Clicking "Post GR" triggers a confirmation dialog and updates the GR status to "Posted", simulating SAP Movement Type 101.

Verify Supplier Invoice: Invoice table displaying base amount, 18% GST tax, and total payable amount. A dedicated 3-Way Match summary panel compares PO value, GR value, and Invoice value. If all three align, the "Verify & Release" button becomes active to release payment, completing the P2P cycle.

4. Screenshots





5. Tech Stack

Layer	Technology	Purpose
UI Framework	OpenUI5 1.108+ (CDN)	SAP's open-source SAPUI5 — all controls, themes, routing
Theme	SAP Horizon (sap_horizon)	Latest SAP Fiori visual design language
Architecture	MVC — XML Views + JS Controllers	Separation of UI, business logic, and data
Data Layer	sap.ui.model.json.JSONModel	Simulates SAP OData V2/V4 service responses
Routing	sap.m.routing.Router	Hash-based SPA navigation between all screens
Layout	sap.ui.layout.Grid / SimpleForm	Responsive breakpoints (XL / L / M / S)
Hosting	GitHub Pages + GitHub Actions	Auto-deploy CI/CD pipeline on push to main branch

6. Unique Points

No SAP System Required: The prototype runs entirely in the browser using OpenUI5 loaded from CDN and a local JSON model, making it fully demonstrable without SAP system access or a SAP Gateway connection.

Authentic SAP Fiori Design Language: All controls used — GenericTile, ObjectListItem, ObjectStatus, StepInput, ActionSheet, SimpleForm — are genuine OpenUI5 components styled with the SAP Horizon theme, identical to what ships in SAP S/4HANA 2023 and above.

Live Two-Way Data Binding: The JSON model is bound reactively across all views. Creating a new PR on the Create screen instantly updates the tile counter on the Launchpad and the PR List table at runtime, demonstrating OpenUI5's live data binding without any page reload.

3-Way Match Financial Control: The Invoice Verification screen implements SAP's core financial control concept — comparing PO value, GR value, and Invoice value before releasing payment. This is the mechanism that prevents companies from paying for goods not received or being overbilled.

Accurate SAP T-Code Mapping: Every screen maps directly to its SAP transaction code equivalent (ME51N, ME28, ME21N, MIGO with Movement Type 101, MIRO), making the prototype directly relatable to a real SAP S/4HANA implementation.

CI/CD Deployment Pipeline: A GitHub Actions workflow automates deployment to GitHub Pages on every push to the main branch, demonstrating awareness of modern DevOps practices alongside SAP Fiori development.

7. Future Improvements

- Connect to a live SAP Gateway OData V4 service by replacing the JSON model with sap.ui.model.odata.v4.ODataModel bound to real SAP S/4HANA backend APIs.
- Implement SAP BTP XSUAA authentication for role-based access control, separating views for Requestor, Approver, Purchaser, and Warehouse Staff roles.
- Add SAP Analytics Cloud embedded charts for spend analysis, vendor performance tracking, and PO cycle time KPIs on the Launchpad.
- Integrate SAP Business Rules Service to automate PR approval routing based on configurable monetary thresholds, replacing manual approvals for low-value requisitions.
- Enable real-time push notifications via SAP Event Mesh to alert approvers when a new PR is raised and notify requestors when their PR is approved or rejected.
- Extend the application to mobile using the SAP Fiori mobile SDK for offline-capable goods receipt posting on warehouse handheld devices.